

Moments

11 problems

4 textbook exercises · 4 strategic practice · 3 homework

Questions below; full solutions at the end of the chapter.

TEXTBOOK EXERCISES

Moment generating functions

Q1 Book Ch.6, Exercise 13

[solution · p. 180](#)

13. (S) A fair die is rolled twice, with outcomes X for the first roll and Y for the second roll. Find the moment generating function $M_{X+Y}(t)$ of $X + Y$ (your answer should be a function of t and can contain unsimplified finite sums).

Q2 Book Ch.6, Exercise 14

[solution · p. 180](#)

14. (S) Let $U_1, U_2, \dots, U_{60}$ be i.i.d. $\text{Unif}(0, 1)$ and $X = U_1 + U_2 + \dots + U_{60}$. Find the MGF of X .

Q3 Book Ch.6, Exercise 20

[solution · p. 180](#)

20. (S) Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X . The *cumulant generating function* is defined to be $g(t) = \log M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the j th *cumulant* of X . Find the j th cumulant of X , for all $j \geq 1$.

21. ⑤ Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

STRATEGIC PRACTICE

2 Moment Generating Functions (MGFs)

Q5 SP 6 §2 · Problem 1

solution · p. 180

1. Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$ using the MGF of X (see also Problem 2 in the Exponential Distribution section).

Q6 SP 6 §2 · Problem 2

solution · p. 181

2. If X has MGF $M(t)$, what is the MGF of $-X$? What is the MGF of $a + bX$, where a and b are constants?

Q7 SP 6 §2 · Problem 3

solution · p. 181

3. Let $U_1, U_2, \dots, U_{60}$ be i.i.d. $\text{Uniform}(0,1)$ and $X = U_1 + U_2 + \dots + U_{60}$. Find the MGF of X .

Q8 SP 6 §2 · Problem 4

solution · p. 181

4. Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X . The *cumulant generating function* is defined to be $g(t) = \ln M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the j th *cumulant* of X . Find the j th cumulant of X , for all $j \geq 1$.

HOMEWORK

Q9 HW 6 · Problem 6

solution · p. 182

6. Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

7. Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$. Then Y has a *Log-Normal distribution* (which means “log is Normal”; note that “log of a Normal” doesn’t make sense since Normals can be negative).

Find the mean and variance of Y using the MGF of X , *without doing any integrals*. Then for $\mu = 0, \sigma = 1$, find the n th moment $E(Y^n)$ (in terms of n).

3. Consider independent Bernoulli trials with probability p of success for each. Let X be the number of failures incurred before getting a total of r successes.

(a) Determine what happens to the distribution of $\frac{p}{1-p}X$ as $p \rightarrow 0$, using MGFs; what is the PDF of the limiting distribution, and its name and parameters if it is one we have studied?

Hint: start by finding the $\text{Geom}(p)$ MGF. Then find the MGF of $\frac{p}{1-p}X$, and use the fact that if the MGFs of r.v.s Y_n converge to the MGF of a r.v. Y , then the CDFs of the Y_n converge to the CDF of Y .

(b) Explain intuitively why the result of (a) makes sense.

CHAPTER 6 · SOLUTIONS

S1 Book Ch.6, Exercise 13

question · p. 177

Solution: Since X and Y are i.i.d., LOTUS gives

$$M_{X+Y}(t) = E(e^{t(X+Y)}) = E(e^{tX})E(e^{tY}) = \left(\frac{1}{6} \sum_{k=1}^6 e^{kt}\right)^2.$$

S2 Book Ch.6, Exercise 14

question · p. 177

Solution: The MGF of U_1 is $E(e^{tU_1}) = \int_0^1 e^{tu} du = \frac{1}{t}(e^t - 1)$ for $t \neq 0$ (and the value is 1 for $t = 0$). Thus, the MGF of X is

$$E(e^{tX}) = E(e^{t(U_1 + \dots + U_{60})}) = \left(E(e^{tU_1})\right)^{60} = \frac{(e^t - 1)^{60}}{t^{60}},$$

for $t \neq 0$ (and the value is 1 for $t = 0$).

S3 Book Ch.6, Exercise 20

question · p. 177

Solution: Using the Taylor series for e^t ,

$$g(t) = \lambda(e^t - 1) = \sum_{j=1}^{\infty} \lambda \frac{t^j}{j!},$$

so $c_j = \lambda$ for all $j \geq 1$.

S4 Book Ch.6, Exercise 21

question · p. 178

Solution: Using the fact that $(1 + x/n)^n \rightarrow e^x$ as $n \rightarrow \infty$ (the compound interest limit, which is reviewed in the math appendix), we have

$$E(e^{tX_n}) = (1 - p_n + p_n e^t)^n = (1 + \lambda(e^t - 1)/n)^n \rightarrow e^{\lambda(e^t - 1)} = E(e^{tX}).$$

S5 SP 6 §2 · Problem 1

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1. Find $E(X^3)$ for $X \sim \text{Expo}(\lambda)$ using the MGF of X (see also Problem 2 in the Exponential Distribution section).

The MGF of an Exponential random variable with rate parameter λ is $M(t) = E(e^{tX}) = (1 - \frac{t}{\lambda})^{-1} = \frac{\lambda}{\lambda - t}$ for $t < \lambda$ (so there is an open interval containing 0 on which $M(t)$ is finite). To get the third moment, we can take the third derivative of the MGF and evaluate at $t = 0$:

$$E(X^3) = \frac{d^3 M(t)}{dt^3} \Big|_{t=0} = \frac{6}{\lambda^3} \left(1 - \frac{t}{\lambda}\right)^{-4} \Big|_{t=0} = \frac{6}{\lambda^3}$$

But a much nicer way to use the MGF here is via pattern recognition: note that $M(t)$ looks like it came from a geometric series:

$$\frac{1}{1 - \frac{t}{\lambda}} = \sum_{n=0}^{\infty} \left(\frac{t}{\lambda}\right)^n = \sum_{n=0}^{\infty} \frac{n!}{\lambda^n} \frac{t^n}{n!},$$

for $|t| < \lambda$. The coefficient of $\frac{t^n}{n!}$ here is the n th moment of X , so we have $E(X^n) = \frac{n!}{\lambda^n}$ for all nonnegative integers n . So again we get $E(X^3) = \frac{6}{\lambda^3}$. This method not only avoids the need to compute the 3rd derivative of $M(t)$ directly, but also it gives us *all* the moments of X in one fell swoop!

S6 SP 6 §2 · Problem 2

question · p. 178

2. If X has MGF $M(t)$, what is the MGF of $-X$? What is the MGF of $a + bX$, where a and b are constants?

The MGF of $-X$ is $E(e^{-tX}) = M(-t)$. The MGF of $a + bX$ is

$$E(e^{t(a+bX)}) = E(e^{at+btX}) = e^{at}E(e^{btX}) = e^{at}M(bt).$$

S7 SP 6 §2 · Problem 3

question · p. 178

3. Let $U_1, U_2, \dots, U_{60}$ be i.i.d. Uniform(0,1) and $X = U_1 + U_2 + \dots + U_{60}$. Find the MGF of X .

The MGF of U_1 is $E(e^{tU_1}) = \int_0^1 e^{tu} du = \frac{1}{t}(e^t - 1)$ for $t \neq 0$, and the MGF of U_1 is 1 for $t = 0$. Thus, the MGF of X is 1 for $t = 0$, and for $t \neq 0$ it is

$$E(e^{tX}) = E(e^{t(U_1 + \dots + U_{60})}) = (E(e^{tU_1}))^{60} = \frac{(e^t - 1)^{60}}{t^{60}}.$$

S8 SP 6 §2 · Problem 4

question · p. 178

4. Let $X \sim \text{Pois}(\lambda)$, and let $M(t)$ be the MGF of X . The *cumulant generating function* is defined to be $g(t) = \ln M(t)$. Expanding $g(t)$ as a Taylor series

$$g(t) = \sum_{j=1}^{\infty} \frac{c_j}{j!} t^j$$

(the sum starts at $j = 1$ because $g(0) = 0$), the coefficient c_j is called the j th *cumulant* of X . Find the j th cumulant of X , for all $j \geq 1$.

Using the Taylor series for e^t ,

$$g(t) = \lambda(e^t - 1) = \sum_{j=1}^{\infty} \frac{\lambda}{j!} t^j$$

so $c_j = \lambda$ for all $j \geq 1$.

6. Let $X_n \sim \text{Bin}(n, p_n)$ for all $n \geq 1$, where np_n is a constant $\lambda > 0$ for all n (so $p_n = \lambda/n$). Let $X \sim \text{Pois}(\lambda)$. Show that the MGF of X_n converges to the MGF of X (this gives another way to see that the $\text{Bin}(n, p)$ distribution can be well-approximated by the $\text{Pois}(\lambda)$ when n is large, p is small, and $\lambda = np$ is moderate).

Using the fact that $(1 + x/n)^n \rightarrow e^x$ as $n \rightarrow \infty$ (which was given in class when the Poisson was first introduced, and is also easy to obtain by taking logs and then using L'Hôpital's Rule), we have

$$E(e^{tX_n}) = (1 - p_n + p_n e^t)^n = (1 + \lambda(e^t - 1)/n)^n \rightarrow e^{\lambda(e^t - 1)} = E(e^{tX}).$$

7. Let $X \sim \mathcal{N}(\mu, \sigma^2)$ and $Y = e^X$. Then Y has a *Log-Normal distribution* (which means “log is Normal”; note that “log of a Normal” doesn't make sense since Normals can be negative).

Find the mean and variance of Y using the MGF of X , *without doing any integrals*. Then for $\mu = 0, \sigma = 1$, find the n th moment $E(Y^n)$ (in terms of n).

The mean and variance of Y can be computed as

$$\begin{aligned} E(Y) &= E(e^X) = e^{\mu + \sigma^2/2} \\ E(Y^2) &= E(e^{2X}) = e^{2\mu + 2\sigma^2} \\ \text{Var}(Y) &= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2} = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1). \end{aligned}$$

The n th moment of Y when $\mu = 0$ and $\sigma = 1$ is

$$E(Y^n) = E(e^{nX}) = e^{n^2/2}.$$

3. Consider independent Bernoulli trials with probability p of success for each. Let X be the number of failures incurred before getting a total of r successes.

(a) Determine what happens to the distribution of $\frac{p}{1-p}X$ as $p \rightarrow 0$, using MGFs; what is the PDF of the limiting distribution, and its name and parameters if it is one we have studied?

Hint: start by finding the $\text{Geom}(p)$ MGF. Then find the MGF of $\frac{p}{1-p}X$, and use the fact that if the MGFs of r.v.s Y_n converge to the MGF of a r.v. Y , then the CDFs of the Y_n converge to the CDF of Y .

Let $q = 1 - p$. For $G \sim \text{Geom}(p)$, the MGF is

$$E(e^{tG}) = p \sum_{k=0}^{\infty} e^{tk} q^k = p \sum_{k=0}^{\infty} (qe^t)^k = \frac{p}{1 - qe^t},$$

for $qe^t < 1$. So the $\text{NBin}(r, p)$ MGF is $\frac{p^r}{(1 - qe^t)^r}$ for $qe^t < 1$. Then the MGF of $\frac{p}{1-p}X$ is

$$E(e^{\frac{tp}{q}X}) = \frac{p^r}{(1 - qe^{tp/q})^r}$$

for $qe^{tp/q} < 1$. Let us first consider the limit for $r = 1$. As $p \rightarrow 0$, the numerator goes to 0 and so does the denominator (since $qe^{tp/q} \rightarrow 1e^0 = 1$). By L'Hôpital's Rule,

$$\lim_{p \rightarrow 0} \frac{p}{1 - (1 - p)e^{tp/(1-p)}} = \lim_{p \rightarrow 0} \frac{1}{e^{tp/(1-p)} - (1 - p)t \left(\frac{1-p+p}{(1-p)^2} \right) e^{tp/(1-p)}} = \frac{1}{1 - t}.$$

So for any fixed $r > 0$, as $p \rightarrow 0$ we have

$$E(e^{\frac{tp}{q}X}) = \frac{p^r}{(1 - qe^{tp/q})^r} \rightarrow \frac{1}{(1 - t)^r}.$$

This is the $\text{Gamma}(r, 1)$ MGF for $t < 1$ (note also that the condition $qe^{tp/q} < 1$ is equivalent to $t < -\frac{1-p}{p} \log(1 - p)$, which converges to the condition $t < 1$ since again by L'Hôpital's Rule, $\frac{-p}{\log(1-p)} \rightarrow 1$). Thus, the scaled Negative Binomial $\frac{p}{1-p}X$ converges to $\text{Gamma}(r, 1)$ in distribution as $p \rightarrow 0$.

(b) Explain intuitively why the result of (a) makes sense.

The result of (a) makes sense intuitively since the Gamma is the continuous analogue of the Negative Binomial, just as the Exponential is the continuous analogue of the Geometric (as discussed in class and seen on the HW 5 problem about *The Winds of Winter*). In fact, that problem essentially shows that if $X \sim \text{Expo}(1)$ then $\lfloor \frac{1}{\log(\frac{1}{1-p})} X \rfloor \sim \text{Geom}(p)$. To convert from discrete to continuous, imagine performing many, many trials where each is performed very, very quickly and has a very, very low chance of success. To balance the rate of trials with the chance of success, we use the scaling $\frac{p}{q}$ since this makes $E(\frac{p}{q}X) = r$, matching the $\text{Gamma}(r, 1)$ mean.